Bequests, Fertility and Barriers to Entrepreneurship:

The Intergenerational Transmission of Economic Inequality

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Abstract

This paper studies how parental fertility and bequest decisions influence children's ability to become entrepreneurs in the presence of financial frictions. Entrepreneurs often face borrowing constraints, which can be alleviated by intergenerational transfers. Empirical evidence shows that high-income parents tend to have fewer children and leave larger bequests, enabling them to allocate more resources per child. As a result, their children face fewer financing barriers to entrepreneurship compared to those from low-income families. However, most existing work on intergenerational transfers and entrepreneurship has emphasized bequests, giving less attention to fertility as a key dimension of parental resource allocation. To address this gap, I develop an overlapping generations model with endogenous fertility, bequest, and occupational choices, calibrated to US data. The model reveals that parental fertility and bequest decisions, when combined with financial frictions, distort children's entrepreneurship choices and reduce efficiency. Counterfactual analysis shows that eliminating financial frictions in the US from the baseline model increases income per capita by 13% but slightly reduces the entrepreneurship rate, raises income inequality, and lowers intergenerational mobility. These findings suggest that financial policy should aim to promote economic development while addressing concerns about income concentration.

Keywords: Entrepreneurship, Financial Frictions, Bequest, Fertility

JEL Classification: O16, E24

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1 Introduction

Entrepreneurship is crucial for economic development, but many entrepreneurs face borrowing constraints that limit their ability to start or grow businesses (Buera, 2008; Evans and Jovanovic, 1989). Intergenerational transfers can help alleviate financial stress, but access to such support is highly unequal. Individuals with high-income parents typically encounter fewer financial barriers, as wealthier households tend to have fewer children (Becker, 1960; Borg, 1989; Docquier, 2004; Jones et al., 2008) and can concentrate more resources and leave larger bequests on each child. These patterns contribute to unequal access to entrepreneurship under financial frictions.

Heterogeneity in family support for entrepreneurship has important implications for efficiency, inequality, and intergenerational mobility. First, the interaction between parental resource allocation and financial frictions can exacerbate the mismatch between entrepreneurial talent and capital. High-ability individuals with limited parental transfers may be unable to access sufficient capital, leading them to either forgo entrepreneurship or operate at suboptimal scale, both of which can reduce aggregate productivity. Second, intergenerational decisions about fertility and bequests shape the next generation's income distribution. While previous research has incorporated bequest behavior in models of entrepreneurship and income distribution, it typically abstracts from fertility. By modeling fertility as an endogenous choice alongside bequests, this paper provides a more comprehensive view of how intergenerational decisions affect the distribution of income and the degree of inequality. Third, disparities in family support affect intergenerational mobility. Because entrepreneurship is a key channel for wealth accumulation and many of the wealthiest individuals are business owners (Cagetti and De Nardi, 2006), children from low-income families face structural barriers that limit their chances of moving to higher positions in the income distribution relative to their parents.

This paper examines how parents' decisions regarding fertility and bequests impact their children's ability to become entrepreneurs when financial markets are imperfect. I build an overlapping generations model where parents choose how many children to have and how much wealth to transfer to each child. These choices determine the financial resources that each child inherits. High-income parents, who face higher child-rearing costs due to greater opportunity costs of their time, tend to have fewer children and leave larger bequests. Consequently, their children can access more startup capital and are more likely to become entrepreneurs when they reach adulthood. At the aggregate level, financial frictions prevent some talented individuals from accessing enough capital. As a result, they either do not start a business or operate below the optimal scale. This misallocation lowers productivity.

I begin with a benchmark model without financial frictions. In each generation, adults make fertility and bequest decisions based on their income and preferences. They derive utility from consumption, raising children, and leaving bequests, following a warm-glow framework. The financial assets inherited by children are the bequests chosen by their parents. In addition, children inherit entrepreneurial ability, which follows an AR(1) process. Individuals live through two periods: childhood and adulthood. During childhood, they consume passively. Upon reaching adulthood, individuals observe their talent and receive their parental bequests. Based on these endowments, they choose to become entrepreneurs or workers and earn income accordingly. In the absence of financial frictions, all entrepreneurs can operate at their optimal scale. After determining their occupation and knowing their income, adults make their own consumption, fertility, and bequest decisions, and the cycle continues across generations.

I next introduce financial frictions to the model arising from the limited enforceability of the credit market contract. Following Buera et al. (2011), individuals who borrow capital from banks can default, retaining a fraction $(1 - s)$ of their profits net of labor costs. To prevent default, banks impose borrowing limits that ensure repayment is more profitable than default for entrepreneurs. These capital limits depend on the individual's assets, entrepreneurial talent, and the bank's ability to seize the profits. If the capital limit exceeds the entrepreneurs' optimal capital level, they operate efficiently. Otherwise, they are credit-constrained and produce at a suboptimal scale. Individuals choose the occupation that yields the highest income, given these constraints. As a result, some highly talented individuals with limited assets either exit entrepreneurship or operate suboptimally, leading to misallocation and inefficiency. As in the benchmark model, individuals then choose consumption, fertility, and the bequest to maximize utility, and their children will inherit the bequests as their assets and inherit the entrepreneurial talent following an AR (1) process. The theoretical model predicts that parental fertility and bequest choices could interact with financial frictions to distort the entrepreneurship decision of the next generation, leading to efficiency loss.

After solving the model, I calibrate the parameters using data from the Panel Study of Income Dynamics (PSID). I begin by estimating individual-level income and consumption profiles over the life cycle through regressions with fixed effects, which allows me to construct expected lifetime income and consumption. I use asset holdings at age 25, when adults are unlikely to accumulate much wealth on their own, as a proxy for bequests received from their parents. Macro-level parameters are calibrated using the General Method of Moments (GMM), targeting the entrepreneurship rate and the income distribution. In the second stage, I use the estimated income from the first stage to calibrate utility parameters by matching the mean of log consumption and the distribution of fertility. The calibration suggests that the U.S. credit market is relatively well functioning, in line with the baseline assumptions in the cross-country studies relating to financial development and productivity(Buera et al. (2011)). At the same time, this paper also helps reconcile the findings in Hurst and Lusardi (2004), who document that entrepreneurial entry is correlated with wealth only at the top of the wealth distribution. Their empirical framework links entry primarily to wealth holdings, abstracting from the interaction between human capital, wealth, and borrowing constraints. By incorporating both human capital and wealth, my model predicts that borrowing constraints in the United States arise mainly for two groups: high-talent individuals with low wealth who operate at a large scale optimally, and low-talent individuals with high wealth who can enter through the leverage effect of wealth.

After the calibration, I conduct counterfactual exercises to evaluate the effects of relaxing financial frictions and shutting down the endogenous fertility channel. Reducing financial frictions, relative to the baseline model calibrated to the US, boosts economic development by enabling highly talented individuals to operate larger firms and absorb more capital. However, this also raises interest rates, which crowds out less competitive entrepreneurs and amplifies returns to existing wealth. As a result, the overall entrepreneurship rate declines slightly, income inequality increases, and intergenerational mobility falls. These findings underscore the importance of designing finance, tax, and transfer policies that account for both economic development and concerns about income concentration.

The counterfactual exercise regarding the fertility channel shows that excluding endogenous fertility decisions, as is common in the existing literature, can lead to overestimation in inequality and mobility under current U.S. financial conditions. This paper makes a key contribution by introducing endogenous fertility decisions into a structural model of entrepreneurship under financial frictions. By modeling parental choices over family size and intergenerational transfers, the analysis captures how these decisions shape children's entrepreneurship outcomes, income distribution, and intergenerational mobility. The results show that omitting fertility behavior can lead to inaccurate measures of inequality and mobility, thereby underscoring the importance of considering the fertility channel in policy evaluations.

The remainder of the paper is structured as follows. Section 2 reviews the related literature and outlines the paper's contribution. Section 3 presents the frictionless benchmark model. Section 4 introduces financial frictions into the framework. Section 5 describes the data and calibration strategy. Section 6 reports the counterfactual analysis. Section 7 concludes.

2 Related Literature

The key mechanism in this paper relates to the literature on fertility-driven dilution of family resources. The classical quantity–quality trade-off literature defines "quality" primarily in terms of children's human capital investment (Becker, 1960; Docquier, 2004) and fertility choices often affect intergenerational mobility and inequality through human capital accumulation and its impact on labor income (Ehrlich and Kim, 2007; Moav, 2005). A related strand of literature emphasizes that fertility also dilutes wealth transfers among children and thereby shapes intergenerational mobility and income inequality through wealth accumulation (Aso, 2024; Cooke et al., 2017; Pestieau, 1984). Complementary work studies how monetary transfers affect wealth and income distributions without explicitly modeling fertility (De Nardi, 2004; De Nardi and Yang, 2016). More recently, scholars have combined fertility choices with both human capital investment and monetary transfers in dynastic frameworks to study how family choices influence social mobility in a more comprehensive way (Daruich and Kozlowski, 2016), though abstracting from the entrepreneurship channel.

However, the influence of intergenerational wealth transmission on mobility and inequality is not limited to the direct effect of inherited wealth, as wealth can have a leverage effect through entrepreneurship. When individuals receive monetary transfers from parents, they can relax borrowing constraints and finance a larger amount of capital when starting a business. Empirically, the relationship between wealth and entrepreneurial entry is nuanced. Though Hurst and Lusardi (2004) argues that wealth holdings only influence the entrepreneurial entry at the top of the wealth distribution, Fairlie and Krashinsky (2012) shows that the irrelevance between business entry and wealth is due to the changing proportion of job losers at different asset levels. After examining the subsamples of job losers and non-job losers separately, they find an increasing rate of entrepreneurial entry along the asset distribution. Taken together, this evidence supports the view that wealth and financial frictions meaningfully shape entrepreneurship.

Entrepreneurship, in turn, is a key driver of wealth concentration and persistence at the top of the income and wealth distributions (Brüggemann, 2021; Quadrini, 1999, 2000; Scheuer, 2014; Terajima, 2006). As a result, the monetary transfers within the family across generations lead to the intergenerational transmission of inequality: individuals from rich families are more likely to overcome entrepreneurial financial barriers and become entrepreneurs, thereby remaining at the top of income and wealth distributions through business ownership.

Existing quantitative work captures the influence of bequest on entrepreneurship and, consequently, income and wealth distributions, but typically abstracts from the endogenous fertility channel. Cagetti and De Nardi (2006) models intergenerational transmission through discount factors but does not explicitly include endogenous bequests. Ivanova (2023) examines how family monetary transfers alleviate financial barriers to entrepreneurship and shape wealth and income distributions in a dynamic model, but it omits the dilution effect of fertility on family resources. However, empirical studies show that family structure and size can affect entrepreneurship (Vladasel, 2023), aligning with the dilution mechanism in family resource allocation.

The main contribution of this paper is to integrate these strands of research by studying how fertility-driven dilution of family monetary transfers interacts with credit constraints to shape entrepreneurship choices and how these dynamics map into intergenerational mobility and the distributions of income and wealth through entrepreneurship's leverage effect.

3 Frictionless model

I first build a model without financial frictions as the benchmark. I adopt a discrete-time OLG model where people experience two stages of life: child and adult. The adults make occupational choices and maximize their utility. The children consume passively.

3.1 Utility maximization

After knowing their income, parents choose consumption c_{it}, total bequest to all children m_{it} and the number of children n_{it} to maximize their utility:

$$\max_{\{c_{it},n_{it},m_{it}\}} \left(\theta_1 c_{it}^{1-\frac{1}{\epsilon}} + \theta_2 m_{it}^{1-\frac{1}{\epsilon}} + (1-\theta_1-\theta_2) n_{it}^{1-\frac{1}{\epsilon}}\right)^{\frac{\epsilon}{\epsilon-1}} \tag{1}$$

$$\text{Subject to:} \quad c_{it} + m_{it} = (1-\tau n_{it}) y_{it}$$

Parameter ϵ is the elasticity of substitution between consumption, total bequest, and fertility. θ_1 and θ_2 represent how much people value consumption and bequest respectively. y_{it} is individual i_t's income. τ is the time cost of having one child. To raise n_{it} children, the parent needs to pay $\tau \times n_{it}$ fraction of their income y_{it}.

The solution to the utility maximization problem is listed as follows:

$$c_{it}^* = \frac{1}{1+(\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \tag{2}$$

$$m_{it}^* = \frac{(\frac{\theta_2}{\theta_1})^\epsilon}{1+(\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it} \tag{3}$$

$$n_{it}^* = \frac{(\frac{1-\theta_1-\theta_2}{\theta_1 \tau})^\epsilon}{1+(\frac{\theta_2}{\theta_1})^\epsilon + (\frac{1-\theta_1-\theta_2}{\theta_1 \tau y_{it}})^\epsilon} \times y_{it}^{1-\epsilon} \tag{4}$$

Consumption and bequest are increasing in an individual's income. The monotonicity of the number of children depends on the value of ϵ. When $\epsilon \leq 1$, n_{it} increase in y_{it}. However, if $\epsilon > 1$, n_{it} will first increase in income when $y_{it} < (\frac{1-\theta_1-\theta_2}{\theta_1 \tau})(\frac{1+(\frac{\theta_2}{\theta_1})^\epsilon}{\epsilon-1})^{\frac{1}{\epsilon}}$. As income increases above the threshold, the number of children will decrease in income due to the increased child cost. Figure 1 indicates the relationship between income and the number of children for different values of ϵ.

I define $b_{it} = \frac{m_{it}}{n_{it}}$ as the average bequest, and each child within the family obtains the same amount of bequest b_{it}. When the child grows up and becomes an adult, they will inherit the amount of b_{it} as their asset. In aggregate, for the next generation $t+1$, a fraction of $\frac{n_{it}}{\sum_{i_t}(n_{it})}$ people will receive b_{it} amount of bequests as their assets. Parents' optimal bequest

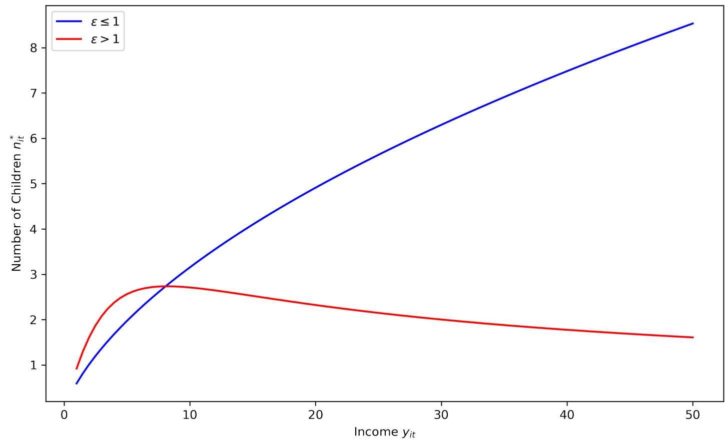

Figure 1: Relationship between Income and Number of Children

and fertility choices determine the distribution of assets for the next generation.

3.2 Individuals' setup

When the children grow up, they inherit bequests from their parents as assets:

$$a_{\text{children},t} = b_{\text{parent},t-1} \tag{5}$$

At the same time, entrepreneurial ability is given by the following equation:

$$\log(z_{\text{children},t}) = \rho \log(z_{\text{parent},t-1}) + \bar{z} + \epsilon_{it} \tag{6}$$

$\epsilon_{it} \sim N(0,1)$. Each individual's talent is partially inherited from the parent and $0 < \rho < 1$. In the long run, the individual-level talent would converge to $e^{\frac{\bar{z}}{1-\rho}}$.

3.3 Occupation

After knowing the asset level a_{it} and entrepreneurial talent z_{it}, the individual i_t first faces the following firm production problem:

$$\max_{\{K_{it},L_{it}\}} \pi_{it} = (A \times z_{it})^{1-\alpha-\beta}(K_{it})^{\alpha} L_{it}^{\beta} - R_t K_{it} - W_t L_{it} \tag{7}$$

After solving the previous problem, the individual knows their optimal capital level, labor level, as well as the profit:

$$K_{it}^{*} = (\frac{\alpha}{R_t})^{\frac{1-\beta}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} A z_{it} \tag{8}$$

$$L_{it}^{*} = (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{1-\alpha}{1-\alpha-\beta}} A z_{it} \tag{9}$$

$$\pi_{it}^{*} = A z_{it} (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} (1-\alpha-\beta) \tag{10}$$

The depreciation rate of capital is assumed to be 1.

As an entrepreneur, the individual can also earn interest on the asset from the bank. Consequently, the total income of an entrepreneur should be:

$$y_{it}^{\mathrm{E}} = \pi_{it}^{*} + R_t a_{it} \tag{11}$$

On the other hand, an individual provides one unit of labor inelastically as a worker. Workers are matched with the entrepreneurs randomly, and each worker earns the average expected wage in the labor market. The income of a worker is:

$$y_{it}^{\mathrm{W}} = W_t + R_t a_{it} \tag{12}$$

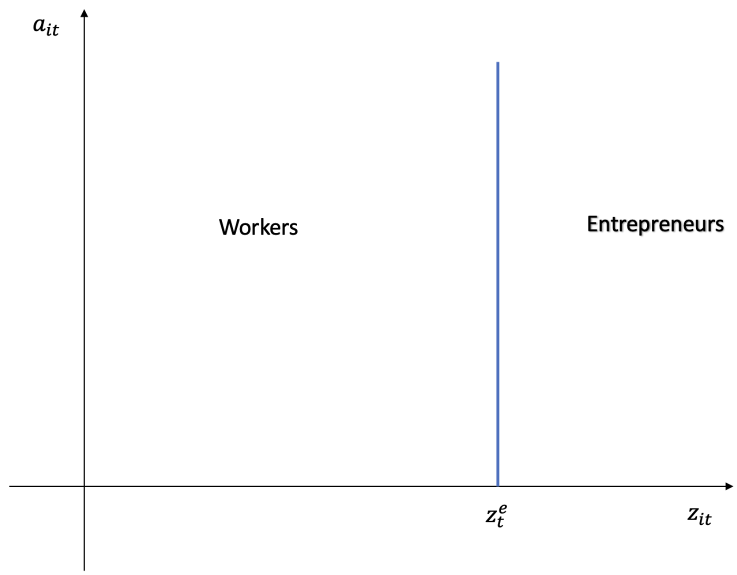

Figure 2: Occupation Choices in the Frictionless Model

By comparing y_{it}^{E} and y_{it}^{W}, the individual makes an occupation choice which is summarized in equation (13). An individual chooses to become an entrepreneur if and only if $y_{it}^{\mathrm{E}} \geq y_{it}^{\mathrm{W}}$. After solving the occupation problem, I get the entrepreneurial cutoff z_t^e in equation (14).

$$o_{it} = \begin{cases} 1 & \text{if} \quad y_{it}^{\mathrm{E}} \geq y_{it}^{\mathrm{W}} \\ 0 & \text{if} \quad y_{it}^{\mathrm{E}} < y_{it}^{\mathrm{W}} \end{cases} \tag{13}$$

$$z_t^e = \frac{1}{A} \frac{\beta}{1-\alpha-\beta} (\frac{R_t}{\alpha})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{W_t}{\beta})^{\frac{1-\alpha}{1-\alpha-\beta}} \tag{14}$$

I ignore the subscript i here because the cutoff doesn't depend on individual-level characteristics. Individuals with entrepreneurial talent higher than z_{it}^e choose to become entrepreneurs with income y_{it}^E. Those beneath the threshold will become a worker with income y_{it}^W. Picture 2 indicates individuals' occupation choices in a frictionless model.

3.4 Equilibrium

Let $\mathbf{x} = (a, z)$ be the state variable in the economy. Following the solutions to the utility maximization problem, the decision rules for occupation and income, and the AR(1) process for entrepreneurial talent, I can calculate the transition function $p(\mathbf{x}'|\mathbf{x})$, which governs the conditional probability of the next state based on the current state.

The stationary equilibrium is defined by a stable distribution $d^*(\mathbf{x})$ of individuals over state variables $\mathbf{x}$; optimal decision rules $c(\mathbf{x})$, $n(\mathbf{x})$ and $m(\mathbf{x})$; the occupational choice $o(\mathbf{x})$; entrepreneurs' optimal solutions $K(\mathbf{x})$ and $L(\mathbf{x})$; interest rate R, wage W. In equilibrium, the following conditions must be satisfied:

(a) Adults maximize utility subject to budget constraints.
(b) Entrepreneurs maximize profits.
(c) Individual makes the optimal occupational choices based on (13).
(d) Labor and capital markets clear. The labor supplied by all workers should be equal to all labor employed by entrepreneurs. The total assets equal all capital employed by entrepreneurs:

$$\int_{i_t \in W} di_t = \int_{i_t \in E} L^*_{it}\, di_t \tag{15}$$

$$\int_{i_t} a_{it}\, di_t = \int_{i_t \in E} K^*_{it}\, di_t \tag{16}$$

(e) d^* is stable.

4 Model with Financial Frictions

The structure of the model with financial frictions is similar to the benchmark model. Parents solve the utility maximization problem in (1) by choosing the optimal consumption, bequest, and fertility choices. When children grow up, they inherit the bequest from their parents as the asset, following equation (5), and get entrepreneurial talent, as specified in equation (6). The only difference is that now entrepreneurs can either default on the contract or pay back the interest to the bank. The bank will set a limit on the capital it will lend to each individual to prevent default.

4.1 Bank and credit market

After knowing their asset and entrepreneurial talent, individuals go to the bank and assess their financial situations. On the one hand, individuals save their money in the bank to earn interest. On the other hand, they can borrow money from the bank to start a firm. However, the credit market is not perfect due to the imperfect enforceability of contracts. Following Buera et al. (2011), I assume that individuals can renege on their contract and keep a $1-s$ fraction of revenue net of labor cost. The only punishment is that they will lose the assets deposited in the bank.

The bank earns zero profit. The interest rate the bank pays is the same as the interest rate the bank lends. To further simplify the model, I assume people put all their assets in the bank and borrow the capital to start a firm. Under the model's assumption, this is equivalent to the case that adults use their assets to start a firm and save the rest in the bank if they have excess assets or borrow from the bank if they don't have sufficient capital.

If $s \geq \frac{\alpha}{1-\beta}$, the economy doesn't experience any financial friction, and the credit market is perfect, we are back to the benchmark model. When $0 \leq s < \frac{\alpha}{1-\beta}$, being aware that individuals can default, the bank will set up a capital limit K_{it}^{limit} to prevent default. The bank wants to ensure that when utilized capital is smaller than or equal to K_{it}^{limit}, the default profit is always not larger than the profit of non-default. Mathematically, the bank sets up K_{it}^{limit} for each individual i_t such that:

$$\begin{aligned} &\max_{\{K_{it}\in[0,K_{it}^{limit}],L_{it}\}} (1-s)\{(Az_{it})^{1-\alpha-\beta}(K_{it})^{\alpha}L_{it}^{\beta} - W_t L_{it}\} - a_{it} \\ &\leq \max_{\{K_{it}\in[0,K_{it}^{limit}],L_{it}\}} \{(Az_{it})^{1-\alpha-\beta}(K_{it})^{\alpha}L_{it}^{\beta} - R_t K_{it} - W_t L_{it}\} + R_t a_{it} \end{aligned} \tag{17}$$

To make it easier to compare, I define $\tilde{s} \in [0,1]$ where $s = \tilde{s} \times \frac{\alpha}{1-\beta}$.

Figure 3 and figure 4 illustrate the capital limit for unbounded and bounded individuals separately. In both figures, the bank wants to set up K_{it}^{limit} such that the default profit is not larger than the production profit. The highest level of K_{it}^{limit} would be given by the

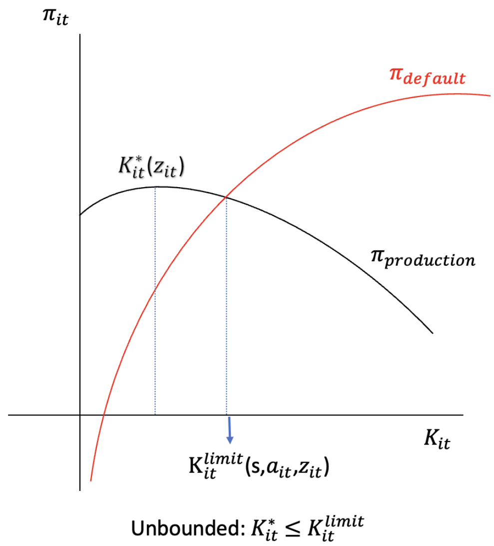

Figure 3: Capital Limit for Unbounded Individual

intersection of π_{default} and $\pi_{\text{production}}$. In figure 3, the capital limit that the bank is willing to lend is higher than the individual's optimal capital level given by equation (8). So this individual will not be bound by the bank, and they can produce at the optimal capital level. In this case, I assume that the highest amount of capital the bank will lend would be K^*_{it} to simplify the model. However, in figure 3, we can notice that $K^{limit}_{it} < K^*_{it}$. In this case, the capital individuals can borrow is less than the optimal amount they need. They can only produce at a suboptimal capital level if they become entrepreneurs. Individuals are unbounded if and only if:

$$\frac{a_{it}}{z_{it}} \geq A \times \frac{\alpha - s(1-\beta)}{1+R_t} (\frac{\alpha}{R_t})^{\frac{\alpha}{1-\alpha-\beta}} (\frac{\beta}{W_t})^{\frac{\beta}{1-\alpha-\beta}} \tag{18}$$

In equilibrium, no one will default. What's more, K^{limit}_{it} is a function of the bank's seizure rate s, personal assets a_{it} and entrepreneurial ability z_{it}. $K^{limit}_{it}(s, a_{it}, z_{it})$ is increasing in s, a_{it} and z_{it}. $K^{limit}_{it}(s, a_{it}, z_{it})$ is concave in a_{it} and z_{it}.

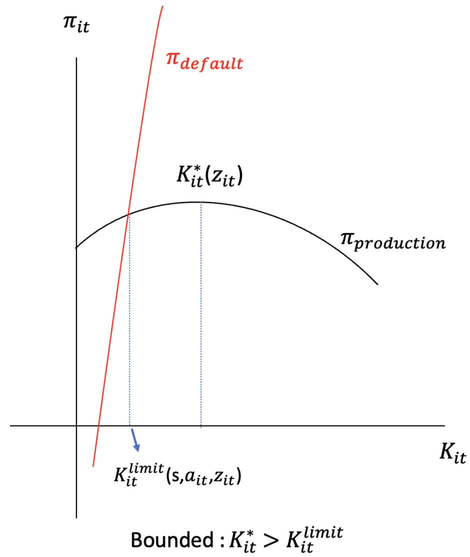

Figure 4: Capital Limit for Bounded Individual

4.2 Occupation, income, and utility

Individuals choose their occupations after knowing their capital limit K_{it}^{limit} from the bank. They first face the following production problem with capital limit K_{it}^{limit}

$$\max_{\{K_{it}\le K_{it}^{limit},L_{it}\}} \pi_{it} = (Az_{it})^{1-\alpha-\beta}(K_{it})^{\alpha}L_{it}^{\beta} - R_tK_{it} - W_tL_{it} \tag{19}$$

For unbounded people, they will produce at the optimal capital level K_{it}^* based on equation (8) and earn profit π_{it}^* given in equation (10). For bounded individuals, they can only produce at a suboptimal level K_{it}^{limit}. Based on inequality (17), an individual's capital limit K_{it}^{limit} is given implicitly by equation (20). The production profit of bounded entrepreneurs would be (21).

$$s(1-\beta)(Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}}(K_{it}^{limit})^{\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}} - RK_{it}^{limit} + (1+R_t)a_{it} = 0 \tag{20}$$

$$\pi_{it}^B = (Az_{it})^{\frac{1-\alpha-\beta}{1-\beta}}K_{it}^{limit\,\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}(1-\beta) - R_tK_{it}^{limit} \tag{21}$$

As in the benchmark model, entrepreneurs can earn interest on their assets from the bank in addition to production profits. As a result, the total income of an entrepreneur would be:

$$y_{it}^{\mathrm{E}} = \begin{cases} y_{it}^{\mathrm{UE}} = \pi_{it}^{*} + R_t a_{it} & \text{if unbounded} \\ y_{it}^{\mathrm{BE}} = \pi_{it}^{B} + R_t a_{it} & \text{if bounded} \end{cases} \tag{22}$$

The worker's income is the same as the benchmark model given by equation (12). Individuals choose the occupation that earns the highest income following equation (13).

For unbounded individuals, the entrepreneurial cutoff is the same as the benchmark model given by equation (14). For bounded individuals, their entrepreneurial cutoff will be decided by equation (23). Bounded individuals with entrepreneurial ability higher than or equal to z_{it}^{BE} become bounded entrepreneurs.

$$z_{it}^{BE} = \frac{1}{A}\left(\frac{W_t + R_t K_{it}^{limit}}{(1-\beta)(K_{it}^{limit})^{\frac{\alpha}{1-\beta}}(\frac{\beta}{W_t})^{\frac{\beta}{1-\beta}}}\right)^{\frac{1-\beta}{1-\alpha-\beta}} \tag{23}$$

Figure 5 shows individuals' occupational choices under financial frictions. The Y-axis shows the asset level, while the X-axis indicates the entrepreneurial talent. The orange line indicates the bound status given by equation (18). Individuals above the line are unbounded, while those under the line are bounded. The vertical blue line is the entrepreneurial cutoff for unbounded individuals given in equation (14). Unbounded individuals with talent equal to or higher than the cutoff will become entrepreneurs. The green curve line demonstrates the entrepreneurial cutoff for bounded people according to equation (23). Bounded people above the line choose to become bounded entrepreneurs.

One comparative static analysis we could do here is to check the influence of the bank's seizure rate s while fixing R and W. In a model without financial frictions, individuals with lower entrepreneurial ability choose to become workers because the labor income is higher than entrepreneurial income: $y_{it}^{W} > y_{it}^{UE}$. When we add financial frictions, we know that bounded entrepreneurial income must be no greater than unbounded entrepreneurial

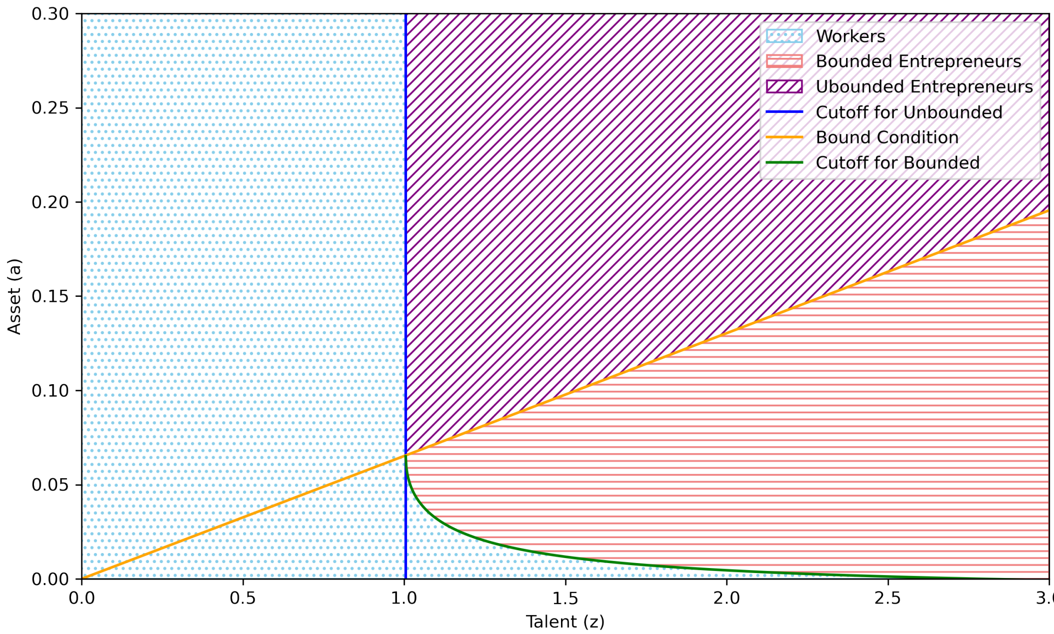

Figure 5: Occupation Choices in the Model with Frictions

income: $y_{it}^{UE} \geq y_{it}^{BE}$. As a result, we should have $y_{it}^{W} > y_{it}^{UE} \geq y_{it}^{BE}$. Individuals left to the unbounded entrepreneurial cutoff still choose to become workers in a world with financial frictions. For those who are entrepreneurs in a frictionless world, there are three situations. Those with small amount of assets now become workers since $y_{it}^{UE} \geq y_{it}^{W} > y_{it}^{BE}$. Individuals with middle-level of assets now become bounded entrepreneurs because $y_{it}^{UE} \geq y_{it}^{BE} > y_{it}^{W}$. Finally, individuals with the highest level of assets remain unbounded entrepreneurs. Comparing the frictionless model with the model considering financial frictions, we can see occupation distortions and mismatches between entrepreneurial talent and capital.

After knowing the occupation and corresponding income, an individual maximizes their utility according to (1). The optimal decision rules are precisely the same as those in the benchmark model listed in equation (2), equation (3), and equation (4).

4.3 Equilibrium

We have $\mathbf{x} = (a, z)$ as the state variable. We can calculate $P(\mathbf{x}'|\mathbf{x})$ as the transition function. In equilibrium, given the bank's seizure rate s, we should have a invariant distribution of the state variable $d^*(\mathbf{x})$; optimal solutions to utility maximization problem $c(\mathbf{x})$, $n(\mathbf{x})$ and

$m(\mathbf{x})$; occupational choice $o(\mathbf{x})$; bound status of individuals $B(\mathbf{x})$ and corresponding capital and labor usage $K(\mathbf{x})$ and $L(\mathbf{x})$; interest rate R and wage W. In equilibrium, we should have:

(a) Adults maximize utility subject to the budget constraint.
(b) The bank sets up capital limits for each individual solving (17).
(c) Entrepreneurs maximize profits given bound status and K_{it}^{limit}.
(d) Adults make optimal occupational choices based on (13).
(e) Labor and capital markets clear:

$$\int_{i_t \in W} di_t = \int_{i_t \in E\&Unbounded} L_{it}^* \, di_t + \int_{i_t \in E\&Bounded} L_{it}^{limit} \, di_t \tag{24}$$

$$\int_{i_t} a_{it} \, di_t = \int_{i_t \in E\&Unbounded} K_{it}^* \, di_t + \int_{i_t \in E\&Bounded} K_{it}^{limit} \, di_t \tag{25}$$

(f) d^* is stable.

5 Calibration

In this section, I use PSID (The Panel Study of Income Dynamics) data to calibrate the model's parameters under financial frictions. I will first describe the calibration plan and then show more details about the calibration method and results.

5.1 Calibration Summary

The inputs of the calibration would be the asset and talent distributions. I estimate these distributions based on the PSID dataset. After knowing the talent and asset distributions, I will first calibrate four macro-level parameters: the TFP shifter A, the capital share α, the labor share β, and the seizure rate s. The targeted moments would be the entrepreneurship rate, the top 5% income share, the log mean of income, and the log median of income.

After calibrating the macro-level parameters, I will use the estimated parameter values and the distribution of talent and assets to generate the estimated income. Then I will calibrate the utility parameters using the estimated income to target the log mean of consumption and the histogram of fertility. The estimated utility parameters includes: consumption weight θ_1, bequest weight θ_2, elasiticity ϵ and time cost per child τ.

5.2 PSID Data

PSID is longitudinal data from a representative sample of U.S. households. It includes intergenerational data on income, consumption, fertility, and occupation over a long period. Before 1999, the survey was conducted annually. After the 1999 wave, it switched to a biennial cycle. My sample consists of the 1968 to 2011 waves of the PSID and keeps the data for the household head. However, the consumption data regarding detailed categories is not included until the 1999 wave. To extend the dataset, I use the consumption data from Attanasio and Pistaferri (2014), which uses socioeconomic variables, prices, and food consumption, which is consistently available in the PSID, to impute the consumption data before 1999.

One concern is that although PSID provides income and consumption data at the year level, some years may be missing. It may not encompass an individual's income and consumption data throughout their life cycle. Therefore, I estimate the income and consumption data for one particular year using the following equation. For y_{it} and c_{it}:

$$\ln y_{it} = \beta_0 + \beta_1 \times age_{it} + \omega_i^y \tag{26}$$

$$\ln c_{it} = \gamma_0 + \gamma_1 \times age_{it} + \omega_i^c \tag{27}$$

t indicates the year when the individuals are aged between 25 and 65. ω_i^y and ω_i^c are the individual fixed effects for income and consumption, which also capture cohort fixed effects. After running the regression, I store the individual fixed effects and estimate yearly income and consumption using the regression coefficients. To get the lifetime income, I use a constant interest rate $R = 5\%$ to discount all the yearly incomes from age 25 to age 65. I use the same method to calculate the expected lifetime consumption.

Figure 6 shows the estimated and real log income levels. Similarly, Figure 7 indicates the relationship between the estimated and real log consumption.

In my model, the asset level refers to the bequest inherited from the parent. Since the bequest data from the PSID contains a lot of missing values, I define the inherited asset as

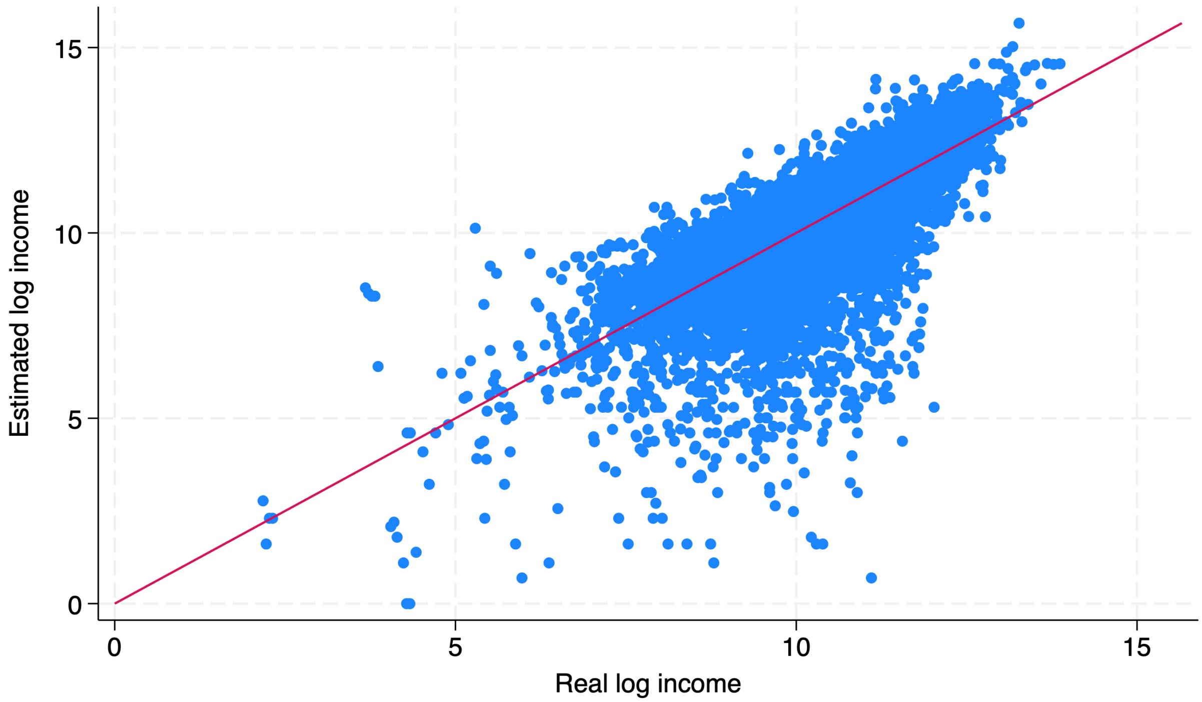

Figure 6: Real log income and estimated log income

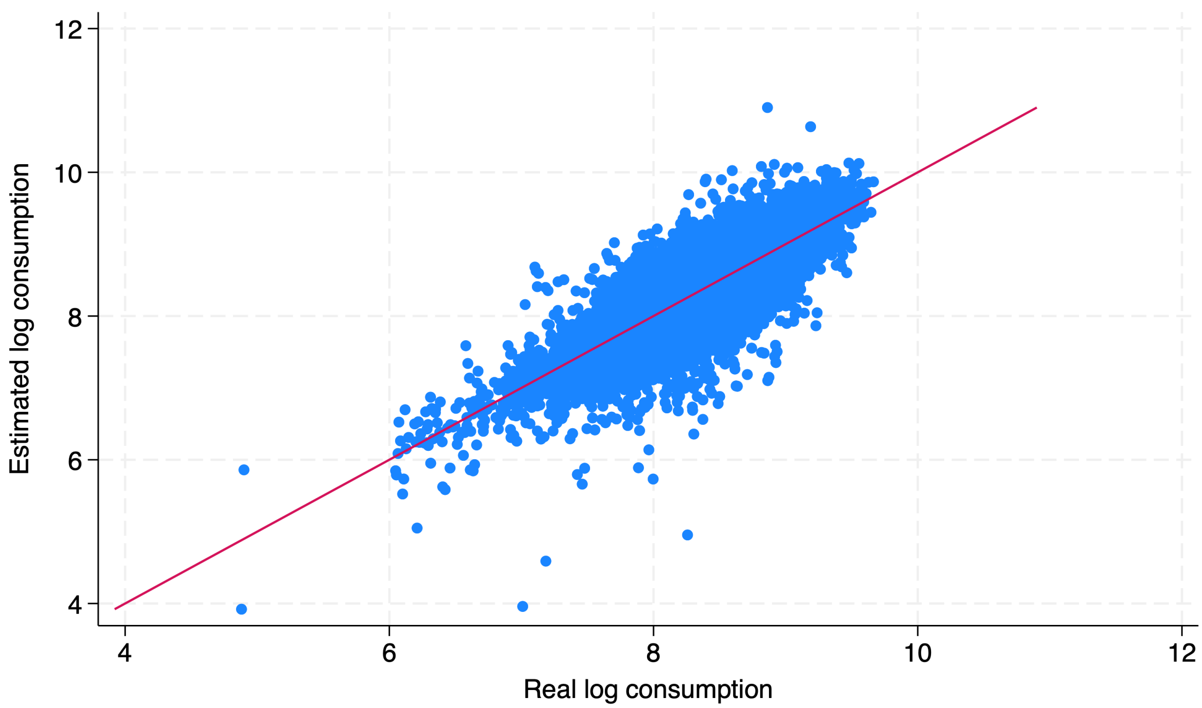

Figure 7: Real log consumption and estimated log consumption.

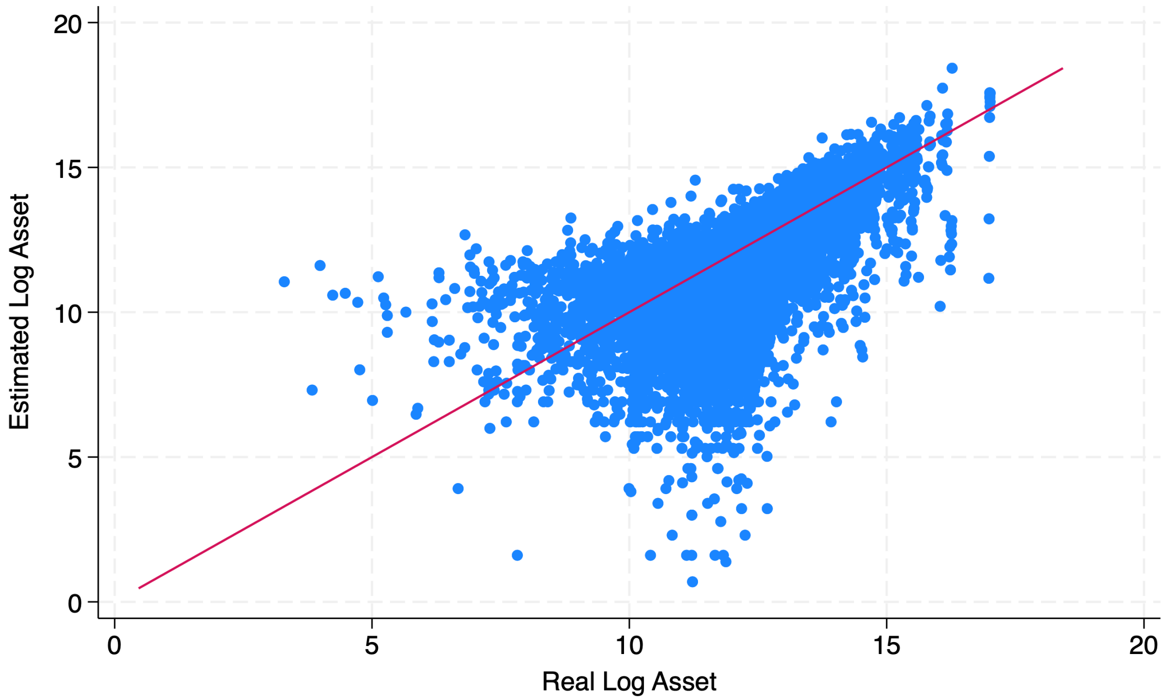

Figure 8: Real log asset and estimated log asset

the asset level at age 25 and estimate it using the following equation:

$$a_{it} = \sigma_0 + \sigma_1 \times \text{age} + \omega_i^a \tag{28}$$

ω_i^a indicates the individual fixed effect, which also captures the cohort fixed effect. Figure 8 shows the relationship between estimated and real log asset levels. After calculating the lifetime income, lifetime consumption, and asset level at age 25, I discount all the values back to the year 1980 using the interest rate $R = 5\%$.

Another important point to discuss is how to identify entrepreneurs in the dataset. I use two variables in the PSID dataset. The first variable relates to whether the household head is reported as self-employed. This question could capture whether the individual is working for someone else or themselves. The second variable regards whether the family owns a business or has a financial interest in any business enterprise. This question provides information about the ownership of a business. If the answers to both questions are yes, I define the individual as an entrepreneur in that particular year. If a person is an entrepreneur for two

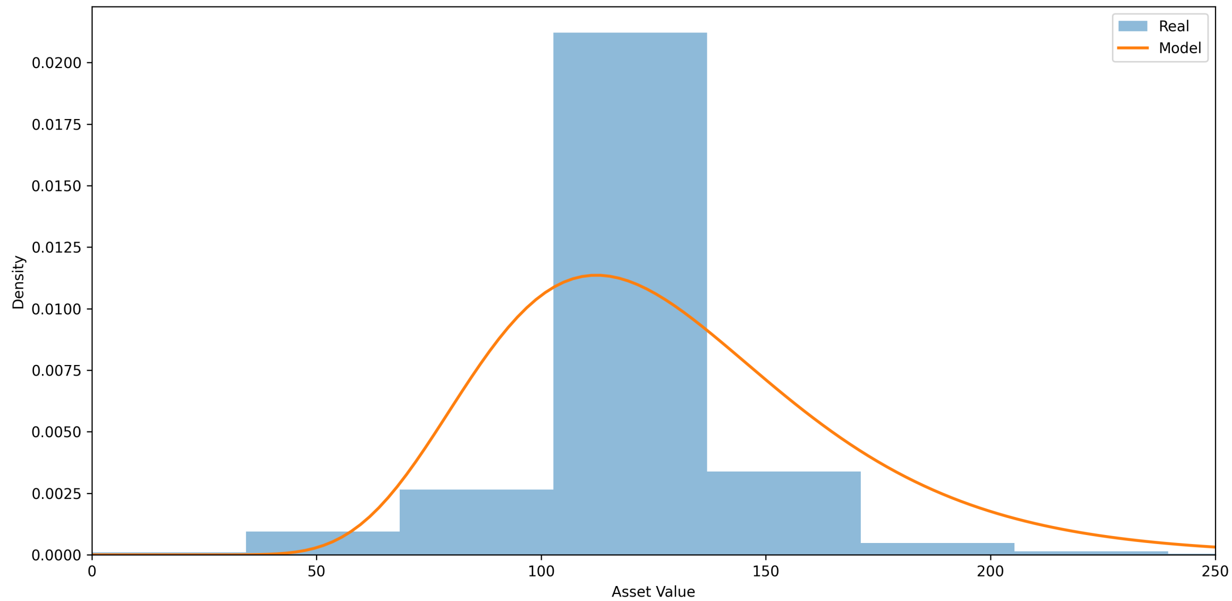

Figure 9: Asset Distribution (in units of $100,000)

consecutive years and for at least half of the years observed in the PSID dataset, I define them as an entrepreneur.

5.3 Asset and Talent Distributions

Following the literature, I assume the asset follows the lognormal distribution. The asset distribution is based on the estimated asset level at age 25 based on the PSID dataset. Figure 9 shows the estimated and real asset distributions.

As for the talent distribution, I first use the following equation to get the residues of lifetime income as the talent measure:

$$\begin{aligned}
\ln(\mathrm{y}_i) \; = \; & \omega_0 + \boldsymbol{\omega}_1^\top \,\textbf{asset_quantile}_i + \boldsymbol{\omega}_2^\top \,\textbf{race}_i + \boldsymbol{\omega}_3^\top \,\textbf{disability}_i \\
& + \boldsymbol{\omega}_4^\top \,\textbf{gender}_i + \boldsymbol{\omega}_5^\top \,\textbf{birthyear}_i \\
& + \boldsymbol{\omega}_6^\top \,(\textbf{asset_quantile}_i \odot \textbf{race}_i) + \boldsymbol{\omega}_7^\top \,(\textbf{asset_quantile}_i \odot \textbf{disability}_i) \\
& + \boldsymbol{\omega}_8^\top \,(\textbf{asset_quantile}_i \odot \textbf{gender}_i) + \boldsymbol{\omega}_9^\top \,(\textbf{asset_quantile}_i \odot \textbf{birthyear}_i) + \varepsilon_i
\end{aligned} \tag{29}$$

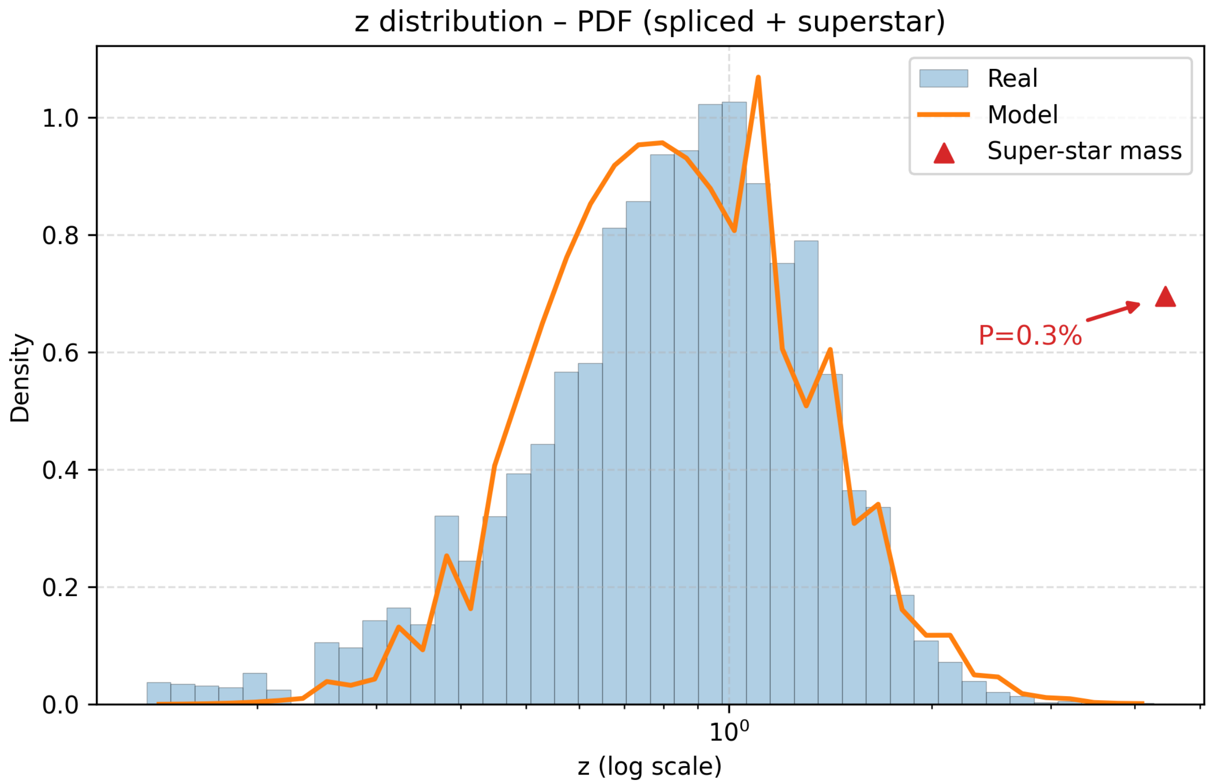

Figure 10: Talent distribution

In the equation, **asset_quantile**$_i$ indicates the wealth quantile of the individual in the society, and **disability**$_i$ indicates whether the individual is handicapped. I treat the residue as the log value of the talent.

I need to estimate both the talent distribution and the AR(1) process parameters. For the initial distribution, I use the log normal distribution to simulate the left tail, use the Pareto distribution to simulate the right tail, and assume there exists a small fraction of super-talented individuals on the far right end. Figure 10 shows the comparison between the real data and simulated talent distribution.

I assume the intergenerational talent transmission follows the AR(1) process. Since the dataset only includes observations for the household head, and most of the households are male, I link the relationship between the father and the children and run the regression following equation (30). The dataset contains 746 matched father–child pairs.

Parameter	Value	Description
ρ	0.29	Persistence
$\overline{z}$	-0.1	Mean of log talent
$\delta_{\epsilon_{it}}$	0.43	Std. dev. of shock

Table 1: Estimated parameters for talent inheritance

$$\ln z_{it+1} = \overline{z} + \rho \ln z_{it} + \epsilon_{it} \tag{30}$$

After running the regression in equation (30), I get the parameter accordingly. Table 1 lists the estimated parameters for the AR(1) process of z.

5.4 Macro parameters

After estimating the distribution of assets and talent, I have four parameters to be estimated at the macro level: the TFP shifter A, the capital share α, the labor share β, and the bank's seizure rate s. The four main targets are the fraction of entrepreneurs within the U.S., the top 5% income share, the log mean of income, and the log median of income.

5.5 Utility parameters

After calibrating the macro-level parameters, I use the estimated income to calibrate the following four parameters in the utility maximization problem: the weight of consumption θ_1, the weight of total bequest θ_2, the elasticity of substitution ϵ, and the time cost per child τ.

From the solution to the utility maximization problem in (1), we know the optimal decisions for consumption and fertility. I use GMM (General Method of Moments) to target the following moments: the log mean of consumption and the distribution of the number of children.

Parameter	Calibrated Value	Description
$\tilde{s}$	0.98	Standardized seizure rate
A	136	TFP shifter
α	0.62	Capital share
β	0.28	Labor share
θ_1	0.13	Consumption weight
θ_2	0.22	Bequest weight
ϵ	4.49	Elasticity
τ	0.19	Per-child time cost

Table 2: Calibrated Parameters

5.6 Calibration results

Table 2 shows the calibrated parameter values. For the macro-level parameters, the standardized bank seizure rate $\tilde{s}$ is very close to 1, indicating that the US's financial market is near perfect. The result indicates that most entrepreneurs in the US are not bound by the bank unless they have high entrepreneurial talent and need to open very large firms.

As for the utility parameters, the consumption weight θ_1 is around 0.13 and the total bequest weight θ_2 is around 0.22. We can observe that the elasticity $\epsilon = 4.49$, which is larger than 1, indicating that the number of children and the income level exhibit a hump-shaped relationship. The time cost of raising one child is 0.19.

Table 3 lists the comparison between the model's predictions and real data for the targeted moments. For all the targeted moments, the model fits the real data well.

Figure 11 represents the comparison of income between the model's prediction and the real data, while Figure 12 and 13 represent the comparison of consumption and the number of children, respectively. For the income and the number of children, the model predicts the distributions very well. For consumption, even though I have only targeted the log mean of the consumption, the model predicts the overall distribution of consumption very well.

Finally, Table 4 compares the model's predictions with empirical data for untargeted mo-

Moment	Target Value	Achieved Value
Entrepreneurship rate	0.099	0.097
Top 5% share	0.14	0.12
Log mean income	3.55	3.37
Log median income	3.43	3.29
Log mean consumption	0.85	0.62
Histogram of n	See Figure 13	

Table 3: Targeted Moments

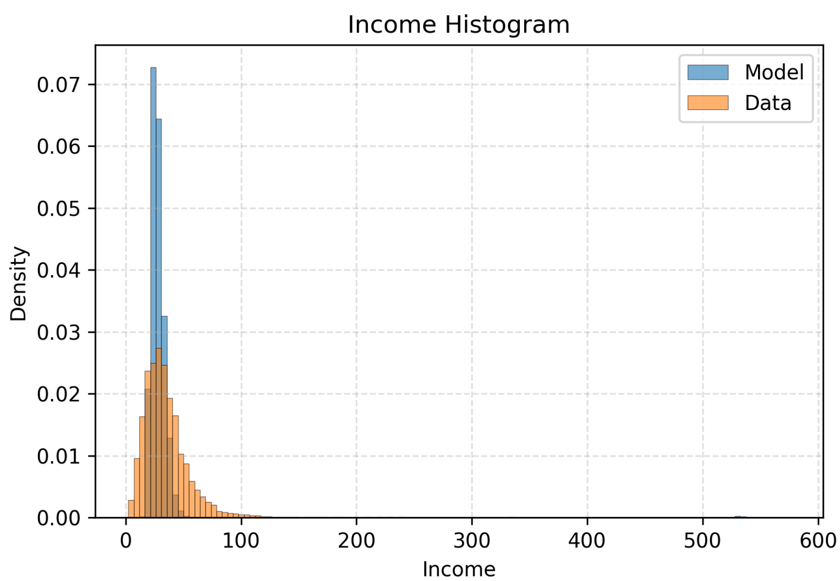

Figure 11: Income: Model vs Data

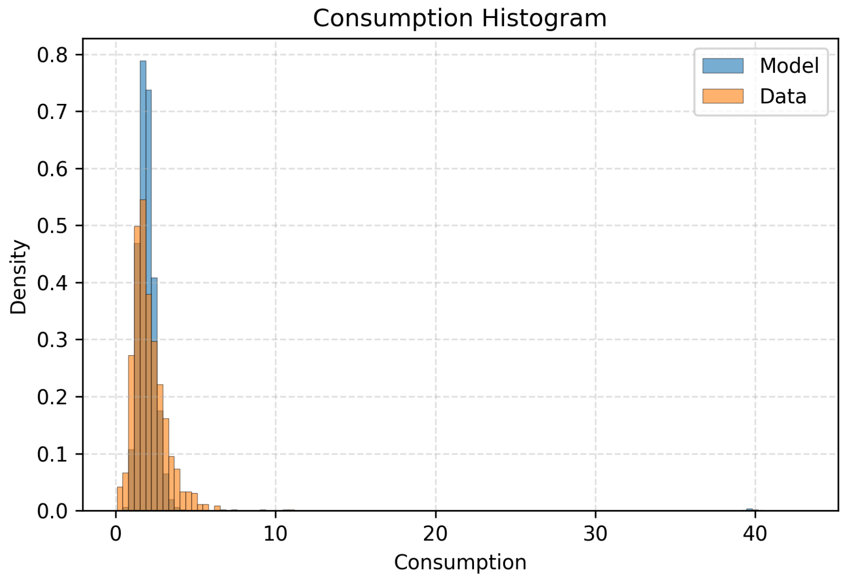

Figure 12: Consumption: Model vs Data

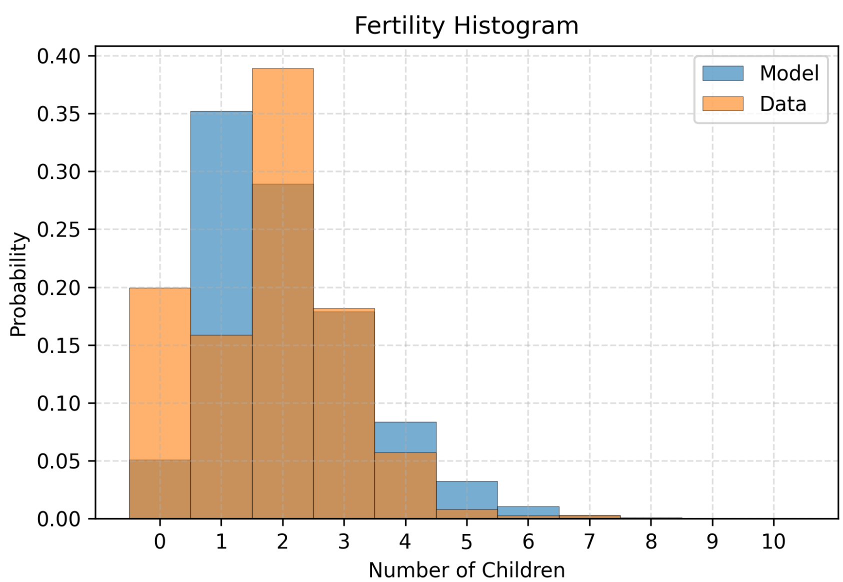

Figure 13: Fertility: Model vs Data (Model predictions round to integers)

Variable	Model	Data
entrepreneurship ratio for the low-income	0.053	0.057
entrepreneurship ratio for the middle-income	0.088	0.073
entrepreneurship ratio for the high-income	0.15	0.168
entrepreneur and worker income ratio	1.661	1.469

Table 4: Untargeted Moments

Entrepreneurship rate	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.216	0.218
$\tilde{s} = 1$	0.200	0.214
$\tilde{s} = 0$	0.114	0.170

Table 5: Counterfactual Analysis for the entrepreneurship rate

ments. The first three rows report entrepreneurship ratios for low-income, middle-income, and high-income individuals. The last row compares the income ratio between entrepreneurs and workers. In all cases, the model's predictions match the empirical data well.

6 Counterfactual Analysis

In this section, I present the counterfactual analysis for changing the financial frictions and shutting down the fertility channel. I want to examine how the entrepreneurship rate, income inequality, mobility, economic development, and social welfare change in the steady state under different conditions.

Table 5 shows the results for the entrepreneurship rate. The table shows that both an increase and a decrease in financial frictions relative to the baseline model would decrease the entrepreneurship rate in the steady state. The difference between Table 5 and Table 3 is because Table 5 indicates the entrepreneurship rate in the long-run. What's more, if I shut down the fertility channel, we would overestimate the entrepreneurship rate.

Next, Table 6 and Table 7 show the results for the Gini Coefficient and the income top

The Gini coefficient	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.560	0.623
$\tilde{s} = 1$	0.609	0.627
$\tilde{s} = 0$	0.643	0.634

Table 6: Counterfactual Analysis for the Gini Coefficient

The top 5% income share	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.555	0.632
$\tilde{s} = 1$	0.594	0.637
$\tilde{s} = 0$	0.630	0.641

Table 7: Counterfactual Analysis for the Top 5% income share

5% share. They are both indicators of income inequality. We can see similar patterns in both tables. Compared with the baseline model, both an increase and a decrease in financial frictions would lead to greater income inequality. Shutting down the fertility channel would overestimate income inequality, except in the case that when $\tilde{s} = 0$ and forcing $n = 1$ would underestimate the Gini coefficient.

Table 8 presents the result for the mobility, which is the probability that a child's income quantile is higher than that of the parent. If I increase or decrease the financial frictions from the baseline model, we would see a drop in mobility. If I shut down the fertility channel, we would see an overestimation of mobility, except when $\tilde{s} = 0$.

It may seem counterintuitive that, when the financial market improves relative to the

Mobility	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.119	0.167
$\tilde{s} = 1$	0.115	0.152
$\tilde{s} = 0$	0.0009	0.0002

Table 8: Counterfactual Analysis for mobility

Income per capita	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.457	0.962
$\tilde{s} = 1$	0.523	0.976
$\tilde{s} = 0$	0.391	0.267

Table 9: Counterfactual Analysis for income per capita

Consumption	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.012	0.043
$\tilde{s} = 1$	0.013	0.044
$\tilde{s} = 0$	0.009	0.012

Table 10: Counterfactual Analysis for consumption

baseline model, entrepreneurship drops slightly and equality and mobility decrease. The mechanism is that reducing the financial barriers from an already well-functioning financial market allows super-talented individuals to absorb more capital and drive up interest rates. Consequently, less competitive firms exit the market, and the return to wealth would increase, leading to fewer entrepreneurs, higher inequality, and lower mobility.

Next, Table 9, Table 10, and Table 11 present the results for income per capita, consumption, and utility, respectively. Decreasing financial frictions would increase income per capita, consumption, and utility. If I force $n = 1$, we would overestimate consumption and utility. As for income per capita, forcing $n = 1$ would overestimate it for the baseline model and for $\tilde{s} = 1$. However, in the collateral case when $\tilde{s} = 0$, excluding the fertility channel would underestimate income per capita.

Finally, I want to check the influence of financial frictions on fertility. Table 12 shows the results. Reducing financial frictions would increase fertility.

7 Conclusion

This paper examines how income-driven parental choices regarding fertility and bequests impact children's entry to entrepreneurship under financial frictions. To study these dynam-

Utility	endogenous fertility	$n = 1$
$\tilde{s} = s^{baseline}$	0.204	0.686
$\tilde{s} = 1$	0.236	0.687
$\tilde{s} = 0$	0.184	0.625

Table 11: Counterfactual Analysis for utility

Fertility	endogenous fertility
$\tilde{s} = s^{baseline}$	0.299
$\tilde{s} = 1$	0.345
$\tilde{s} = 0$	0.274

Table 12: Counterfactual Analysis for fertility

ics, I propose an OLG model with endogenous fertility, bequest, and occupational choices, and calibrate the model using U.S. data. The model indicates that the interaction between parental decisions and financial friction could lead to the misallocation of entrepreneurial talent and capital. Counterfactual analysis indicates that reducing financial frictions relative to the baseline model would increase income per capita but slightly reduce entrepreneurship, thereby increasing inequality and reducing mobility. These findings underscore the importance of designing financial policies that promote efficiency while also addressing concerns about income concentration.

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